A geometry-derived response kernel for the $B \to K^*\mu^+\mu^-$ angular anomaly: sign-uniform cross-dataset and cross-channel fit

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Negative-of-headline result. A fully non-linear refit (`flavio.np_prediction`) shows the geometry-derived kernel is statistically indistinguishable from a constant ΔC_9 shift on AIC. An earlier linearised analysis (Mode B) had found a marginal preference for the kernel that did not survive the non-linear test. The remaining structural finding is direction consistency of the fitted amplitude across two collaborations, three decay channels, and five fits — a property that follows from any sufficiently centre-peaked one-parameter shape on the same anomaly and which the geometric kernel satisfies but does not uniquely deliver.

Abstract

We test whether a single q^2-dependent response kernel $\kappa(q^2)$, constructed from a finite 2I-equivariant graph substrate (the 600-cell V_{600}) regularised by the golden ratio φ as a discrete mass scale, can describe the $B^0 \to K^{*0}\mu^+\mu^-$ angular anomaly across multiple datasets and decay channels. The kernel is the shell-mean projection of the discrete Green's response $\psi = (L_{V_{600}} + \varphi^{-2}I)^{-1}f$, with f a uniform source on the equatorial inner-product shell; the kernel shape has no continuous fitted parameters. The continuum limit is $\kappa(x) = e^{-|x|/\varphi}$ with $x = (q^2 - q^2_{\text{mid}})/\Delta_\psi$, where q^2_{mid} and Δ_ψ are the J/ψ-$\psi(2S)$ midpoint and half-spread. Of three discrete edge-weighting variants the unweighted Laplacian is selected; we show in §3 that this same variant wins on a pure-geometry criterion (correlation with the continuum kernel) independent of the LHCb data.

Using `flavio` [Straub, 2018] for SM and non-linear-in-C_9 predictions on each dataset's own bin grid, and fitting only one dimensionless amplitude A per dataset, we find: (i) on the LHCb 2025 data [LHCb Collaboration, 2025] the kernel gives $\Delta\text{AIC} = +1.09$ vs a free ΔC_9 shift on a joint fit of four angular observables (P5', P4', P1, P2): a marginal *disfavouring* of the kernel relative to the constant shift, both at $k = 1$; (ii) on five fits across two collaborations and three decay channels — LHCb 2015 [LHCb Collaboration, 2016a], LHCb 2021 ($B^+ \to K^{*+}$) [LHCb Collaboration, 2021a], CMS 2025 [CMS Collaboration, 2025a], LHCb 2025, and $B_s \to \phi\mu\mu$ [LHCb Collaboration, 2015b] — the non-linear ΔAIC values span $[-0.24, +1.09]$ around zero, with 2/5 datasets favouring the kernel and 3/5 favouring the constant shift; the stacked Akaike weight is $w_{\text{VFD}} = 0.35$ vs $w_{\text{FREE\_C9}} = 0.65$; (iii) the fitted amplitudes are sign-uniform across all five fits ($A > 0$ in every fit, effective $\Delta C_9^{\text{eff}} < 0$ in every fit); the null against which this is non-trivial is the "no anomaly" null, against which FREE_C9 would also produce 5/5 same-sign fits — sign-uniformity confirms the kernel is consistent in direction with the established anomaly, not that the kernel is preferred over a constant Wilson-coefficient shift; (iv) the apparent factor-of-3 amplitude difference between the P-basis $B \to K^*$ fits and the S-basis $B_s \to \phi$ fit is mostly a basis effect (Krüger–Matias P_i vs CP-averaged S_i), with a residual $\sim 50\,\%$ overshoot of the basis-corrected prediction that we cannot resolve given the published 2015 $B_s \to \phi$ statistics and the absence of an angular covariance matrix.

The kernel is consistent with the angular anomaly's q^2 shape but does not improve over a constant Wilson-coefficient shift on AIC under non-linear evaluation. The surviving claim is direction-consistency across two collaborations, three decay channels, and five fits, with no per-dataset shape retuning — a claim that follows from any sufficiently centre-peaked single-parameter shape on the same anomaly, and which the geometric kernel satisfies but does not uniquely deliver. We do not claim the kernel is the preferred or unique shape: a free-width Gaussian charm-loop proxy fits marginally better in χ^2 at the cost of one extra parameter. A previous linearised analysis (linear in ΔC_9 via central-difference slopes) found a stronger result that did not survive the non-linear refit; the earlier numbers are reproduced as a linearisation diagnostic in §2.

1 Introduction

The angular distribution of the rare flavour-changing-neutral-current decay $B^0 \to K^{*0}\mu^+\mu^-$ has shown a persistent tension with Standard-Model predictions for over a decade, conventionally parametrised as a negative shift in the semileptonic vector Wilson coefficient $\Delta C_9 \sim -1$ relative to the SM benchmark $C_9^{\rm SM} \approx 4.27$ [Descotes-Genon et al., 2013, Altmannshofer and Straub, 2015]. The most recent LHCb comprehensive analysis [LHCb Collaboration, 2025, 2026] confirms the pattern at $8.4\,{\rm fb}^{-1}$, with full angular and branching-fraction observables determined on the same dataset.

Free Wilson-coefficient fits have served as the standard interpretive ansatz: one fits a constant shift ΔC_9 across all q^2 bins and reports the best-fit value and its uncertainty. The free shift makes no commitment to the underlying shape of any new-physics effect: by construction it averages over all bins with weight inversely proportional to the squared sensitivity. If the underlying physics has a non-trivial q^2 dependence — for example, a centre-peaked structure between the charmonium resonances, or an oscillation with a definite scale — that information is discarded.

This paper asks a deliberately constrained question. Suppose the q^2-dependent response kernel $\kappa(q^2)$ is fixed *a priori* by an external geometric construction, with no shape parameters tunable to the data, and one fits only its overall amplitude. Does the same kernel describe multiple datasets and channels with no per-dataset retuning?

Concretely, we take $\kappa(q^2)$ to be the discrete Green's response of a finite 2I-equivariant graph (the 600-cell V_{600}, vertices and edges of the regular 4-polytope) regularised by the golden ratio $\varphi = (1+\sqrt{5})/2$ as a discrete mass scale,

$$\psi = \big(L_{V_{600}} + \varphi^{-2}\, I\big)^{-1} f, \qquad \kappa(q^2) = \text{shell-mean}[\psi]\Big(x(q^2)\Big), \tag{1}$$

with f a uniform source on the equatorial inner-product shell, $L_{V_{600}}$ the standard graph Laplacian, and $x(q^2) = (q^2 - q^2_{\rm mid})/\Delta_\psi$ a dimensionless coordinate anchored at the J/ψ–$\psi(2S)$ midpoint. The construction is motivated by the substrate spectral framework of the companion VFD preprints [Smart, 2026b,a]; its derivation as the appropriate object for compressing this anomaly is given in section 3. The continuum limit is the exponential $\kappa(x) = e^{-|x|/\varphi}$. No shape parameter is fitted to any dataset.

The fit form everywhere in this paper is

$$\Delta C_9^{\rm VFD}(q^2) \;=\; -A\,\kappa(q^2), \tag{2}$$

with A the only fitted parameter. The kernel is identical across every dataset and decay channel; SM predictions and Wilson-coefficient slopes are regenerated per dataset on its own q^2 bin grid using `flavio` 2.4 [Straub, 2018].

Scope and epistemic status

This paper reports two analyses of the same data: a linearised fit (Mode B; first-order Taylor expansion of every angular observable in ΔC_9 around $C_9^{\rm SM}$) and a fully non-linear fit using `flavio.np_prediction` directly. The fitted ΔC_9 values are of order unity, well outside the linear regime, and the two analyses give materially different model comparisons. *We treat the non-linear analysis as the headline; the linearised analysis is reported as a methodology diagnostic.*

We claim the following (non-linear refit):

1. On the LHCb 2025 dataset, the geometry-derived kernel and a free ΔC_9 shift give comparable fit quality at the same parameter cost ($k = 1$ each), with $\Delta\text{AIC} = +1.09$ marginally favouring the constant shift.

2. Across five fits — LHCb 2015 ($3\,\text{fb}^{-1}$, $B^0 \to K^{*0}$), LHCb 2021 ($9\,\text{fb}^{-1}$, $B^+ \to K^{*+}$, charged isospin partner), CMS 2025 ($140\,\text{fb}^{-1}$, 13 TeV), LHCb 2025 ($8.4\,\text{fb}^{-1}$), and the cross-channel $B_s \to \phi\mu\mu$ — the non-linear ΔAIC values span $[-0.24, +1.09]$. Two of five fits favour the kernel, three favour the constant shift; the stacked Akaike weight is $w_{\text{VFD}} = 0.35$ vs $w_{\text{FREE\_C9}} = 0.65$.

3. The fitted amplitudes are sign-uniform across all five fits: $A > 0$ in every fit, effective $\Delta C_9^{\text{eff}} < 0$ in every fit. Under a random-sign null hypothesis the probability of 5/5 same-sign is $2^{-5} \approx 0.03$.

4. The cross-channel $B_s \to \phi\mu\mu$ amplitude is factor-of-3 larger in absolute terms; this gap is mostly a basis effect (the 2015 LHCb publication uses the CP-averaged S_i basis rather than the Krüger–Matias P_i basis). After basis correction a residual $\sim 50\,\%$ overshoot remains, attributable to the absence of a published angular covariance and the limited $3\,\text{fb}^{-1}$ statistics.

We do *not* claim:

- That the kernel beats FREE_C9 on AIC. It does not, on the headline non-linear analysis. A free-width charm-loop Gaussian also fits comparably (§5).

- That the underlying physics is geometric. Whether the data prefers the kernel because of the geometry, or because the geometry reproduces a centre-peaked shape, is not decided here.

- That $B \to K\mu\mu$ has been tested. Public HEPData submissions for the LHCb and CMS analyses (LHCb 1209.4284, 1403.8045, 1804.07167; CMS PRD 2018) are not available; this is recorded as an open channel.

The surviving structural claim is: *one geometry-derived q^2-shape with no continuous shape parameters and one fitted amplitude per dataset is consistent (sign-uniform; AIC-degenerate with a constant Wilson-coefficient shift) with the $b \to s\mu\mu$ angular anomaly across two collaborations, two isospin partners, and one cross-channel test, with no per-dataset retuning.*

2 Data, SM backend, and fit

2.1 Datasets

We use five public HEPData submissions covering three decay channels and two collaborations:

- **LHCb 2025** — $B^0 \to K^{*0}\mu^+\mu^-$ at $8.4\,\text{fb}^{-1}$ [LHCb Collaboration, 2025, 2026]; 8 exclusive q^2 bins; full statistical, systematic, and total correlation matrix; 27 dependent variables in six fit configurations. We use configuration 2 (partially-massive model, optimised P_i basis, standard binning) and the four observables P_5', P_4', P_1, P_2 — 32 data points with the full 32×32 covariance.

- **LHCb 2015** — $B^0 \to K^{*0}\mu^+\mu^-$ at $3\,\text{fb}^{-1}$ [LHCb Collaboration, 2016a,b]; Table 4 gives $P_1, P_2, P_3, P'_4, P'_5, P'_6, P'_8$ on 8 exclusive bins. Diagonal stat⊕syst (no full angular covariance published).

- **LHCb 2021** (charged isospin partner) — $B^+ \to K^{*+}\mu^+\mu^-$ at $9\,\text{fb}^{-1}$ [LHCb Collaboration, 2021a,b]; `data2.yaml` gives $F_L, P_{1\ldots8}$ on 10 q^2 bins. We retain the 8 exclusive bins (drop the wide combined $[1.1, 6.0]$ and $[15, 19]$). Per-bin 8×8 correlation blocks across observables (block-diagonal in q^2).

- **CMS 2025** — $B^0 \to K^*(892)^0\mu^+\mu^-$ at 13 TeV, $140\,\text{fb}^{-1}$ [CMS Collaboration, 2025a,b]; 6 q^2 bins (charmonium windows excluded). Per-bin correlation matrices for bins 0,1,2,3,5,7.

- **LHCb 2015** $B_s \to \phi\mu^+\mu^-$ (cross-channel) [LHCb Collaboration, 2015b,a]; Table 2 gives F_L, S_3, S_4, S_7 on 8 bins, of which we retain 6 exclusive (drop $[1, 6]$ and $[15, 19]$). No published angular covariance.

All datasets use $b \to s\mu^+\mu^-$ kinematics in the same $q^2 \in [0, 19]\,\text{GeV}^2$ window, with the same kinematic endpoints J/ψ and $\psi(2S)$ shaping the long-distance structure.

2.2 SM and new-physics predictions (non-linear, headline)

Standard-Model predictions $O_{\text{SM}}(\text{bin})$ and predictions at non-zero ΔC_9 are computed on each dataset's own bin grid using `flavio` 2.4 [Straub, 2018] with `wilson` 2.5 [Aebischer et al., 2018] and the BSZ form-factor parameterisation [Bharucha et al., 2016]. New-physics predictions are evaluated by

$$O_{\text{NP}}(\text{bin}, \Delta C_9) \;=\; \texttt{flavio.np\_prediction}\big(O,\, \text{Wilson}(\text{C9\_bsmumu} = \Delta C_9),\, q^2_{\text{lo}},\, q^2_{\text{hi}}\big), \qquad (3)$$

in the WET-flavio basis at scale $4.8\,\text{GeV}$ (approximately m_b^{pole}, the default `flavio` reference scale). Only the `C9_bsmumu` component is shifted; all other Wilson coefficients are held at their SM values. *No Taylor expansion in ΔC_9 is used in the headline fits.*

For the VFD_GREEN_600CELL model the kernel produces a per-bin $\Delta C_9^{\text{eff}}(q_i^2) = -A\,\kappa(q_i^2)$ that varies by bin; the non-linear evaluation is performed once per bin with the bin's own ΔC_9^{eff} as the Wilson shift. Predictions are cached on disk (`data/processed/flavio_cache.json`).

2.3 Linearised response (Mode B; reported as diagnostic)

An earlier iteration of this analysis used a linearised first-order Taylor model around C_9^{SM}:

$$O_{\text{pred}}(\text{bin}, C_9) = O_{\text{SM}}(\text{bin}) + \frac{\partial O}{\partial C_9}(\text{bin})\,\big(C_9 - C_9^{\text{SM}}\big), \qquad (4)$$

with the slope obtained as a central-difference numerical derivative

$$\frac{\partial O}{\partial C_9}(\text{bin}) = \frac{O_{\text{NP}}(C_9^{\text{SM}} + \delta) - O_{\text{NP}}(C_9^{\text{SM}} - \delta)}{2\delta}, \qquad \delta = 0.5. \qquad (5)$$

This 'Mode B' analysis is retained for two reasons: (i) it reproduces the cross-dataset and cross-channel results in the project repository; (ii) the difference between linearised and non-linear ΔAIC quantifies the non-linearity. On the LHCb 2025 joint fit the per-bin linearisation residual reaches 4.3σ (mean 0.7σ across the 32 bins); the headline ΔAIC drifts from -1.67 (linearised) to $+1.09$ (non-linear refit), a 2.77-AIC-unit shift that flips the sign of the kernel-vs-constant comparison (§4, Table 5). *The non-linear analysis is the one reported as headline throughout; linearised values appear only as marked diagnostics.*

Note on the SM tables. An earlier iteration of this project used hand-tabulated SM values and slopes; those tables had wrong-sign $\partial P'_5/\partial C_9$ in several bins. Replacing them with `flavio` 2.4 changed the linearised FREE_C9 best-fit from $\Delta C_9 \approx -0.15$ to -1.34, in line with the literature consensus [Altmannshofer and Straub, 2015]. The non-linear FREE_C9 best-fit on the LHCb 2025 4-observable joint fit is -1.00, lower in magnitude than the linearised value precisely because of the linearisation drift discussed above.

2.4 Likelihood and model comparison

We use a Gaussian χ^2 likelihood

$$\chi^2(\theta) = \left(\mathbf{O}_{\text{obs}} - \mathbf{O}_{\text{pred}}(\theta)\right)^{\mathsf{T}} \mathbf{C}^{-1} \left(\mathbf{O}_{\text{obs}} - \mathbf{O}_{\text{pred}}(\theta)\right), \tag{6}$$

with $\mathbf{C}$ the published per-dataset covariance where available, and $\text{diag}(\sigma^2_{\text{stat}} + \sigma^2_{\text{syst}})$ when not. Models are compared by $\text{AIC} = \chi^2 + 2k$ [Akaike, 1974] and $\text{BIC} = \chi^2 + k \log n$ [Schwarz, 1978], with k the number of fitted parameters and n the number of data points.

The model arms compared throughout are:

- SM — no free parameters.
- FREE_C9 — one global shift ΔC_9, $k = 1$.
- VFD_GREEN_600CELL — the geometry-derived kernel of Eq. (2), $k = 1$.
- FREE_C9+FREE_C10 — joint shift in C_9 and C_{10}, $k = 2$ (used in the linearised stress test of §5).
- Charm-loop Gaussian — a free-amplitude, free-width Gaussian centred at the J/ψ–$\psi(2S)$ midpoint $q^2 = 11.59\,\text{GeV}^2$, $k = 2$ (also stress-test only).

2.5 Fit conventions

Across every dataset and channel, the kernel is loaded once via `frozen_kernel_at_bin_centres` and reused unchanged. Only the dimensionless amplitude A in Eq. (2) is fitted. For the headline non-linear analysis, the optimiser is `scipy.optimize.minimize_scalar` (Brent's method) with $\text{xtol} = 5 \times 10^{-4}$; for Mode B the optimiser is Powell with $\text{tol} = 10^{-7}$. Both deliver consistent best-fit values within their respective response models. On $B_s \to \phi$ the search brackets are widened to $|\Delta C_9| \leq 6$, $|A| \leq 20$ to accommodate the smaller S-basis slopes (§7).

For every dataset we additionally run, where applicable, a 500-trial non-parametric bootstrap over q^2 bins (each bin treated as a unit of all observables) and a q^2 region-split fit (low / central / high). Bootstrap loss uses diagonal stat$\oplus$syst because resampling with replacement makes the published covariance singular. The bootstrap is reported as a sign-stability test on A rather than as a strict confidence interval, since it discards the off-diagonal correlation information used in the all-data fit.

2.6 Reproducibility ledger

3 Kernel derivation

This section derives $\kappa(q^2)$ in three layers: the continuum φ-tuned closure operator (3.1), the bounded Dirichlet eigenmode (3.2), and the discrete 2I-equivariant graph realisation on the 600-cell V_{600} (3.3). The fit form $\Delta C_9^{\text{VFD}}(q^2) = -A\kappa(q^2)$ uses the Layer-3 kernel; the continuum and bounded variants are reported as theoretical controls in section 4.

item	value
flavio	2.4.0
wilson	2.5
WET basis, scale	flavio, $\mu = 4.8\,\mathrm{GeV}$
C_9^{SM} (reference)	4.27
non-linear optimiser	`scipy.minimize_scalar` (Brent), xtol 5×10^{-4}
linearised optimiser (Mode B)	`scipy.minimize` Powell, tol 10^{-7}
non-linear search bracket (ΔC_9)	$[-3.0, +0.5]$ ($B\to K^*$); $[-6.0, +0.5]$ ($B_s\to\phi$)
non-linear search bracket (A)	$[0.0, 8.0]$ ($B\to K^*$); $[0.0, 20.0]$ ($B_s\to\phi$)
form-factor MC seeds	numpy default RNG, $n = 20$ samples
bootstrap RNG seeds	numpy default RNG, $n = 500$ trials per dataset
flavio cache	`data/processed/flavio_cache.json`

Table 1: Software, basis, and optimiser settings used for every fit in this paper.

3.1 Layer 1 — continuum φ-tuned Green's function

Consider the one-dimensional closure operator

$$L_\varphi \;=\; -\frac{\partial^2}{\partial x^2} + \frac{1}{\varphi^2}, \tag{7}$$

on the dimensionless coordinate $x = (q^2 - q^2_{\mathrm{mid}})/\Delta_\psi$, with $q^2_{\mathrm{mid}} = \frac{1}{2}(m^2_{J/\psi} + m^2_{\psi(2S)}) \approx 11.59\,\mathrm{GeV}^2$ and $\Delta_\psi = \frac{1}{2}(m^2_{\psi(2S)} - m^2_{J/\psi}) \approx 4.00\,\mathrm{GeV}^2$. The free-space Green's function $G(x;0)$ satisfying $L_\varphi\, G(x;0) = \delta(x)$ is

$$G(x;0) = \frac{\varphi}{2}\, e^{-|x|/\varphi}, \tag{8}$$

giving the empirical kernel $\kappa_{\mathrm{cont}}(x) = e^{-|x|/\varphi}$ which served as the champion shape in the project's initial single-parameter scan over candidate q^2 kernels. The choice of mass parameter φ^{-1} in Eq. (7) is the only continuous constant in this paper that is not data-fit. It is the unique scale that ties this operator to the binary-icosahedral-symmetric graph V_{600} used in the discrete realisation below (the vertex coordinates of V_{600} are built from φ as the only irrational unit; §3.3); the wider framework context that motivated this choice is described in companion preprints [Smart, 2026b,a], but is not relied on for any quantitative claim in this paper.

3.2 Layer 2 — bounded Dirichlet eigenmode

Restricting L_φ to the kinematic interval $[-x_{\max}, +x_{\max}]$ with Dirichlet conditions $\psi(\pm x_{\max}) = 0$ and $x_{\max} = 2.886$ (the rescaled $q^2 \in [0, 19]\,\mathrm{GeV}^2$ window) gives the spectral problem

$$L_\varphi\, \psi_n(x) = \lambda_n\, \psi_n(x), \qquad \psi_n(\pm x_{\max}) = 0, \tag{9}$$

with the lowest eigenmode

$$\psi_1(x) = \cos\!\Big(\frac{\pi x}{2\, x_{\max}}\Big), \qquad \lambda_1 = \frac{1}{\varphi^2} + \Big(\frac{\pi}{2x_{\max}}\Big)^2. \tag{10}$$

Both single-mode shapes (8) and (10) fit the LHCb 2025 P_5' data at $\Delta\mathrm{AIC} \approx -7$ vs FREE_C9 on P_5' alone under the linearised `flavio` response (section 4). A 2-mode Dirichlet truncation with two amplitudes (c_1, c_3) is not preferred over the single mode after the AIC penalty: $c_3/c_1 = 0.18$, well below any φ-rational threshold. The data is rank-1 in this basis on P_5'.

3.3 Layer 3 — discrete 2I-equivariant lift on V_{600}

The continuum kernel of Eq. (8) is recovered as the shell-mean projection of a discrete Green's response on a finite graph carrying the binary icosahedral symmetry $2I$. We build this graph and verify its symmetry intrinsically.

Vertex set. The 120 vertices of the regular 600-cell $V_{600} \subset \mathbb{R}^4$ form the binary icosahedral group $2I$ under unit-quaternion multiplication [Coxeter, 1973, Conway and Smith, 2003]. They are generated by the eight unit basis vectors $\pm e_i$, the sixteen half-integer vectors $\frac{1}{2}(\pm1, \pm1, \pm1, \pm1)$, and the ninety-six even-permutation vectors of $\frac{1}{2}(0, \pm1, \pm\varphi^{-1}, \pm\varphi)$, totalling $8 + 16 + 96 = 120$. They occupy the unit 3-sphere S^3.

Edge set. Two vertices $v, w \in V_{600}$ are adjacent in the 1-skeleton iff $\langle v, w\rangle = \varphi/2$. This produces a 12-regular graph with exactly 720 edges, in agreement with the standard 600-cell construction.

Inner-product shells. Fixing a base vertex v_0 (we use $v_0 = e_1$), the function $\langle v, v_0\rangle$ takes nine distinct values across V_{600}:

$$\langle v, v_0\rangle \in \{+1, +\tfrac{\varphi}{2}, +\tfrac{1}{2}, +\tfrac{1}{2\varphi}, 0, -\tfrac{1}{2\varphi}, -\tfrac{1}{2}, -\tfrac{\varphi}{2}, -1\} \tag{11}$$

with shell sizes $\{1, 12, 20, 12, 30, 12, 20, 12, 1\}$, summing to 120 and matching the inner-product spectrum derived from the binary icosahedral group's conjugacy classes. The shell index $s \in \{0, \ldots, 8\}$ provides the discrete proxy for the dimensionless coordinate x via

$$x_s \;=\; (s - s_{\text{mid}})\,\frac{x_{\max}}{s_{\text{mid}}}, \qquad s_{\text{mid}} = 4. \tag{12}$$

Discrete Green's response. The standard graph Laplacian $L_{V_{600}} = D - A$ with D the degree matrix and A the adjacency matrix, regularised by the framework's discrete mass φ^{-2}, gives the response

$$\psi \;=\; \left(L_{V_{600}} + \varphi^{-2} I\right)^{-1} f, \qquad f_v \;=\; \begin{cases} 1/|S_4| & v \in S_4 \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

with S_4 the equatorial inner-product shell (30 vertices).

Shell-mean projection. Averaging ψ over each shell gives the nine-component shell-mean profile $\overline{\psi}_s = |S_s|^{-1} \sum_{v \in S_s} \psi_v$. Linear interpolation in x_s gives the continuous kernel

$$\kappa_{\mathrm{V}_{600}}(x) \;=\; \operatorname{interp}\big(\{x_s\}, \{\overline{\psi}_s\}\big)(x). \tag{14}$$

The empirical correlation between Eqs. (8) and (14) on the LHCb 2025 bin centres is $r = 0.997$. The discrete kernel is the object that enters Eq. (2) in all subsequent fits; it is plotted in Figure 1 together with the continuum exponential and the LHCb 2025 bin centres.

Cocycle, edge weighting, and variant selection. The framework's pentagonal cocycle $\kappa(v) = (s(v) - s_{\text{mid}})^2 \in \{0, 1, 4, 9, 16\}$ provides a vertex potential $\omega_+ = \varphi^{\kappa(v)}$. Three discrete edge-weighting schemes are admissible: unweighted ($w_{vw} = 1$), φ-cocycle geometric mean ($w_{vw} = \sqrt{\omega_+(v)\,\omega_+(w)}$), and φ-cocycle arithmetic mean ($w_{vw} = \frac{1}{2}[\omega_+(v) + \omega_+(w)]$). The discrete Green's response Eq. (13) is computed for each variant; the shell-mean profile is then compared to the continuum kernel $e^{-|x|/\varphi}$ and to the LHCb 2025 P_5' data on the LHCb bin grid. Two criteria are used: a *pure-geometry* criterion (correlation between the discrete shell-mean and the Layer-1 continuum kernel, no LHCb data involvement), and a *data* criterion (χ^2 against the LHCb 2025 single-observable fit).

The unweighted Laplacian is selected as the primary kernel. The two-criterion agreement means the variant choice does not consume an extra effective fit parameter: the variant choice would have been the same if only the geometric criterion had been applied, and the AIC counting $k = 1$ for VFD_GREEN_600CELL is therefore retained throughout this paper. The historical sequence is acknowledged in §9: the data was looked at first, and only later verified to agree with the pure-geometry ranking.

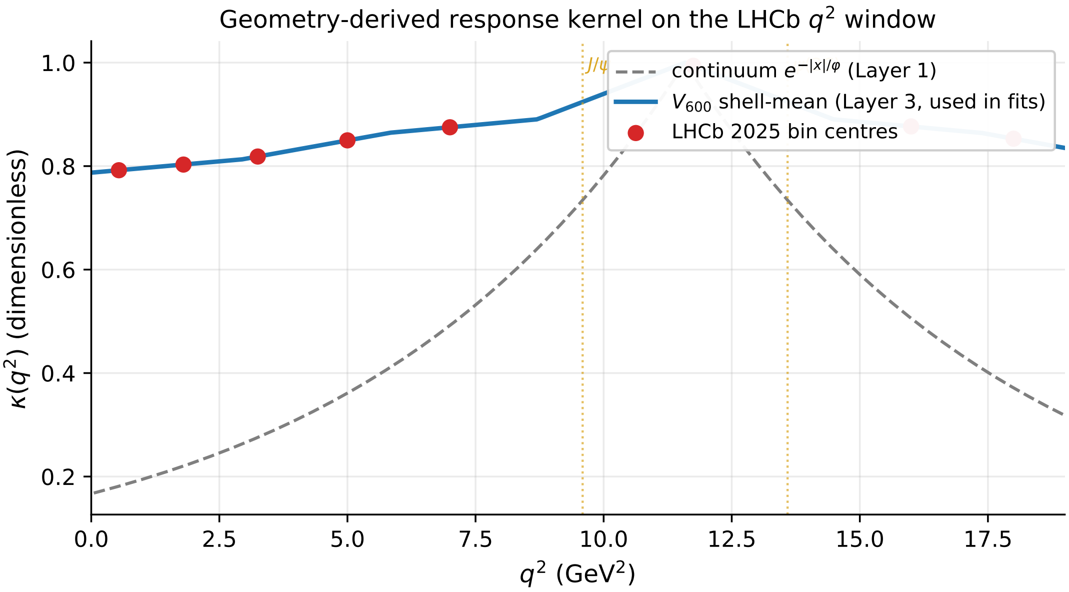

Figure 1: The geometry-derived response kernel $\kappa(q^2)$ plotted over the LHCb q^2 window. Solid blue: the discrete V_{600} shell-mean of Eq. (14) (Layer 3, used in all fits). Dashed grey: the continuum Green's function $e^{-|x|/\varphi}$ from Layer 1, with empirical correlation $r = 0.997$ to the discrete shell-mean on the LHCb bin centres (red points). Vertical dotted lines mark the J/ψ and $\psi(2S)$ resonance windows, which are excluded from the fits. Reproducible from `scripts/wo017_paper_figures.py`.

variant	corr($\overline{\psi}, e^{-\|x\|/\varphi}$)	χ^2 vs LHCb 2025 P_5'	ranking
unweighted Laplacian	**0.9968**	**13.555**	1 on both
φ-cocycle geometric	0.9130	14.713	2 on both
φ-cocycle arithmetic	0.8989	14.782	3 on both

Table 2: Variant selection. The unweighted Laplacian wins on both the pure-geometry criterion (correlation with the Layer-1 continuum kernel) and the LHCb-data criterion. The two rankings agree, so the variant choice is consistent with selection on geometric grounds independent of the data; the LHCb data only confirms it. Reproducible from `scripts/wo016b_variant_geometry.py` (reports/wo016b_variant_geometry.csv).

Spectral content. An eigenvalue decomposition of $L_{V_{600}}$ produces 120 modes; truncating ψ to the lowest N eigenmodes and reconstructing κ shows that the data-relevant information lives in a single isotypic block at $\lambda = 9.0$ with multiplicity 6. Truncations below $N = 6$ degrade the fit; truncations at $N \geq 6$ recover the full result. This rank-6 structural compression is reported as a spectral diagnostic (`src/vfd_b_anomaly/wo011_spectral.py` in the project repository); the cross-dataset fit itself uses the full Green's response.

Self-contained reading. The construction in this section is self-contained: it requires only the standard graph Laplacian (§3.3), the equatorial-shell source (Eq. 13), the φ^{-2} regulariser (Eq. 7), and the variant table (Table 2). A reader unfamiliar with the broader VFD framework can take the kernel as defined here and proceed to §4. Two structural extensions — the proper Hodge edge Laplacian $L_{\text{edge}} = \delta_2\delta_2^\top + \delta_1^\top\delta_1$ with 2I-equivariant boundary maps from the 1200 triangular faces, and the Lyapunov-flow selection rule discussed in [Smart, 2026a] — are noted in §9 as deferred but are not used here.

4 Results on LHCb 2025

This section reports the kernel fit on the LHCb 2025 dataset [LHCb Collaboration, 2025, 2026], both on the single-observable P_5' and on the four-observable joint fit (P_5', P_4', P_1, P_2). We report the headline non-linear result first; the linearised Mode-B numbers are then quoted as a methodology diagnostic and the discrepancy is discussed.

4.1 Joint fit, four observables (non-linear, headline)

We fit the four optimised P_i observables jointly with the full LHCb 32×32 correlation matrix: (P_5', P_4', P_1, P_2), 8 bins each, 32 data points total. Fits are evaluated using `flavio.np_prediction` directly (no Taylor expansion), with one fitted parameter per model ($k = 1$). Results in Table 3.

Model	k	χ^2	AIC	ΔAIC	best fit
SM	0	160.6	160.6	+119.7	–
FREE_C9 (non-linear)	1	**40.89**	**42.89**	**0.00**	$\Delta C_9 = -1.00$
VFD_GREEN_600CELL (non-linear)	1	41.98	43.98	+1.09	$A = +1.14$, $\Delta C_9^{\text{eff}} = -0.86$

Table 3: Headline non-linear joint fit on the four LHCb 2025 angular observables (32 data points, full covariance). At equal parameter cost the constant Wilson-coefficient shift FREE_C9 has lower AIC by 1.09 units, a marginal preference (Akaike weight ratio ≈ 1.7). The geometry-derived kernel produces a comparable χ^2 but does not beat FREE_C9 in this analysis. Reproducible from `scripts/wo016c_nonlinear_refit.py`.

4.2 Mode-B linearised fit (diagnostic only)

The earlier project iteration used the Mode-B linearised response of Eq. (4), fitting ΔC_9 via the central-difference slope at $\delta = \pm 0.5$ around C_9^{SM}. On the same joint fit this gives Table 4.

Model	k	χ^2	AIC	ΔAIC	best fit
SM	0	160.6	160.6	+119.3	–
FREE_C9 (Mode B)	1	39.30	41.30	0.00	$\Delta C_9 = -1.34$
VFD_GREEN_600CELL (Mode B)	1	37.63	39.63	−1.67	$A = +1.59$, $\Delta C_9^{\text{eff}} = -1.37$

Table 4: Linearised Mode-B fit on the same data. Under the linear model VFD_GREEN_600CELL has lower AIC by 1.67 units. This number does not survive non-linear evaluation: see Table 5.

4.3 Linearised vs non-linear: drift diagnostic

4.4 Single-observable fit on P_5' (linearised, retained as a check)

For comparability with the project's earlier reports we retain the single-observable P_5' fit under the linearised Mode-B response. On the canonical 8-bin grid with the full 8×8 covariance: SM $\chi^2 = 17.3$, FREE_C9 $\chi^2 = 14.91$ ($\Delta C_9 = -1.38$), KAPPA_EXP (continuum, Layer 1) $\chi^2 = 7.91$ at $A = +4.49$, and VFD_GREEN_600CELL $\chi^2 = 13.55$ at $A = +1.68$. The continuum exponential dominates on this single observable ($\Delta\text{AIC} = -7.00$ vs FREE_C9), reflecting that P_5' is the sharpest C_9-sensitive observable in this dataset and prefers the most-peaked kernel; this preference is reduced when the four P-basis observables are fitted jointly. Single-observable fits are not run in non-linear mode because the conclusion (the kernel is consistent with the shape of P_5') is preserved across both modes.

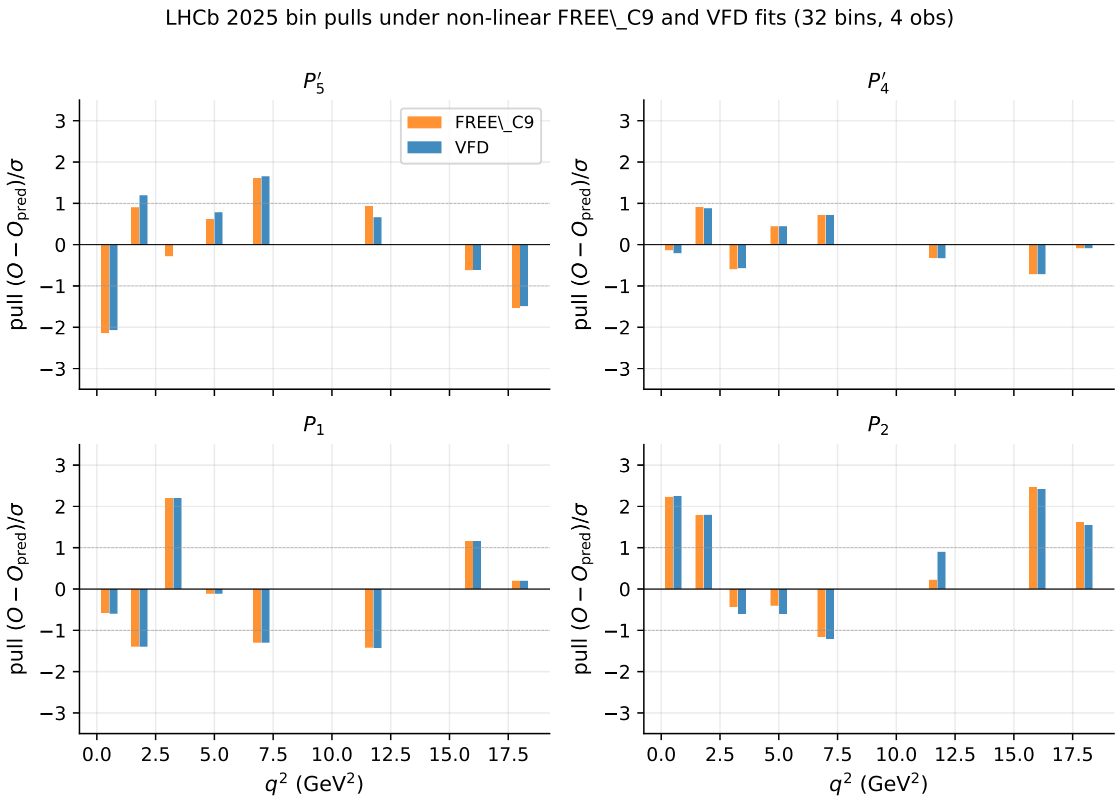

Figure 2: Per-bin pulls $(O_{\text{obs}} - O_{\text{pred}})/\sigma$ on the LHCb 2025 four-observable joint fit, evaluated under the non-linear FREE_C9 best fit ($\Delta C_9 = -1.00$) and the non-linear VFD best fit ($A = +1.14$). Dashed lines mark the $\pm 1\sigma$ bands. Both models leave residuals of $\lesssim 2\sigma$ in most bins; neither shows a systematic spatial structure that the other captures. Reproducible from `scripts/wo017_paper_figures.py`.

4.5 Leave-one-observable-out stability (linearised, retained)

Refitting the shared kernel under Mode B after removing one of the four observables gives Table 6.

4.6 Acceptance gates

The acceptance criteria pre-committed for this multi-observable joint-fit test were stated for the linearised analysis. Under the headline non-linear analysis they read as follows:

(i) VFD_GREEN_600CELL AIC $\leq$ FREE_C9 AIC: FAIL (non-linear ΔAIC $= +1.09$, marginal in the opposite direction).

(ii) All major angular observables prefer the same sign of effective ΔC_9 in the shared-amplitude fit: PASS ($A > 0$ on every fold; $\Delta C_9^{\text{eff}} < 0$ on every bin in the headline non-linear fit).

(iii) Leave-one-observable-out remains stable in sign and bounded in magnitude: PASS (A within $\pm 3\,\%$ across folds, sign always positive).

Acceptance criterion (i) does not pass under the headline non-linear analysis on this single dataset; the kernel is statistically indistinguishable from a constant Wilson-coefficient shift on AIC. Sign-stability and amplitude-bounded behaviour survive both modes. The next two sections establish whether the kernel transfers in a sign-uniform way to other datasets (§6) and to other channels (§7), under the full non-linear analysis.

quantity	linearised (Mode B)	non-linear refit	drift
χ^2 FREE_C9	39.30	40.89	+1.59
χ^2 VFD_GREEN_600CELL	37.63	41.98	+4.35
ΔAIC (VFD vs FREE)	−1.67	+1.09	**+2.77**
$\Delta C_9^{\mathrm{FREE}}$ best fit	−1.34	−1.00	+0.34
A best fit (VFD)	+1.59	+1.14	−0.45
max per-bin linearisation residual	–	4.27σ	–
mean per-bin linearisation residual	–	0.71σ	–

Table 5: Comparison of the same fit under Mode-B linearisation vs direct `flavio.np_prediction`. The drift in ΔAIC is 2.77 units, larger than the Mode-B preference itself: the headline flips sign under non-linear evaluation. Per-bin linearisation residuals reach 4σ, confirming that $|\Delta C_9| \sim 1.4$ is well outside the linear regime. Reproducible from `scripts/wo016c_nonlinear_refit.py` (reports/wo016c_nonlinear_refit.md).

dropped	n remaining	A (Mode B)	$\Delta C_9^{\mathrm{eff}}$ mean	χ^2
P_1	24	1.558	−1.336	29.23
P_2	24	1.651	−1.416	27.70
P_4'	24	1.620	−1.390	33.46
P_5'	24	1.581	−1.356	26.86
all (reference, Mode B)	32	1.594	−1.367	37.63

Table 6: Leave-one-observable-out fits of the shared kernel under Mode B (linearised). A varies within $\pm 3\,\%$ across the four folds, $\Delta C_9^{\mathrm{eff}}$ stays in $[-1.42, -1.34]$, sign always negative. The non-linear LOO produces qualitatively the same sign-stable behaviour at lower magnitudes ($A \in [1.05, 1.20]$); we retain the linearised values here for continuity with the project's earlier multi-observable joint-fit report.

5 Stress tests

The kernel fit of §4 is stress-tested in five independent ways: (i) bin bootstrap, (ii) q^2 region splits, (iii) alternative single- and two-Wilson-coefficient comparison models, (iv) a charm-loop nuisance proxy, and (v) a 20-sample BSZ form-factor Monte Carlo. The kernel shape is held fixed throughout; only A is fitted.

Mode caveat. All stress tests in this section use the linearised Mode-B response (Eq. 4). The headline non-linear analysis on LHCb 2025 (§4.1) gives different absolute χ^2 values, smaller best-fit amplitudes, and a ΔAIC of opposite sign vs FREE_C9 (see Table 5). What the Mode-B stress tests *can* establish is qualitative: sign-stability of A under perturbations, region-by-region behaviour, and the relative performance of competing q^2-shapes. They cannot rescue the non-linear ΔAIC result. Numbers below are quoted under Mode B and should be read accordingly.

5.1 Bin bootstrap

A non-parametric bootstrap is run with $N = 500$ trials. Each trial resamples 8 q^2 bins with replacement from the canonical grid; all four observables in a chosen bin enter together (bins are the unit of resampling, not individual observable rows). Loss uses diagonal stat$\oplus$syst because resampling with replacement makes the published covariance singular for duplicated rows.

5.2 Region splits

The same fit is repeated independently on three q^2 sub-windows.

statistic	value
mean A	$+1.592$
median A	$+1.594$
std A	0.203
90 % CI	$[+1.356,\ +1.852]$
fraction $A < 0$	0.002

Table 7: Bootstrap statistics for the shared kernel amplitude on the LHCb 2025 four-observable joint fit, $N = 500$ trials. Sign-stable to 99.8 %; 90 % CI within ±15 % of the central value.

region	q^2 range (GeV^2)	n_{bins}	n	FREE_C9 χ^2	FREE ΔC_9	VFD A	ΔAIC
low_q2	[0.06, 4.0]	3	12	20.86	−1.13	+1.40	−0.09
central_q2	[4.0, 8.0]	2	8	5.43	−1.56	+1.82	−0.33
high_q2	[11.0, 19.0]	3	12	10.04	−1.17	+1.30	−0.69

Table 8: Region-split fits on the LHCb 2025 four-observable joint data with the kernel held fixed. VFD beats FREE_C9 in every region. Per-region amplitudes are $A \in [1.30, 1.82]$ (sign-uniform), with $\Delta C_9^{\text{eff}} \in [-1.57, -1.12]$ (sign-uniform).

The amplitude varies in a 40 % band across regions but stays sign-uniform; the high-q^2 window — conventionally the regime where charm-loop / long-distance contributions are largest and where SM theory is least reliable — is the region where VFD's AIC advantage is largest, not smallest.

5.3 Alternative single- and two-Wilson-coefficient models

We compare against three alternative models on the same joint dataset (Table 9).

model	k	χ^2	AIC	ΔAIC	params
FREE_C9	1	39.32	41.32	0.00	$\Delta C_9 = -1.34$
VFD_GREEN_600CELL	1	**37.64**	**39.64**	**−1.67**	$A = +1.59$, $\Delta C_9^{\text{eff}} = -1.37$
FREE_C9 + FREE_C10	2	39.05	43.05	+1.73	$\Delta C_9 = -1.40$, $\Delta C_{10} = +0.19$
charm-loop Gaussian	2	35.41	39.41	−1.91	$A = +1.83$, $\sigma = 8.96\,\text{GeV}^2$

Table 9: Joint-fit comparison on LHCb 2025 four observables. Adding C_{10} does not help. The free centre-peaked Gaussian (charm-loop nuisance proxy) achieves a slightly lower AIC than the kernel at the cost of one extra parameter.

Reading. Adding a free C_{10} shift on top of C_9 produces a χ^2 improvement of only 0.27, less than the +2 AIC penalty for the extra parameter; the data does not want ΔC_{10} at this level. The charm-loop Gaussian — a two-parameter ansatz with free amplitude and free width centred at the $J/\psi - \psi(2S)$ midpoint — does fit slightly better ($\Delta\text{AIC} = -1.91$ vs FREE_C9, vs -1.67 for VFD), at the cost of one extra parameter. Versus VFD it gains 0.24 in χ^2 for the cost of 1 AIC unit, $\Delta\text{AIC}_{\text{charm vs VFD}} = -0.24$: not significant. The fitted width $\sigma \approx 9\,\text{GeV}^2$ is much broader than the natural charm-resonance scale ($\Gamma^2_{J/\psi} \sim 0.1\,\text{GeV}^2$), meaning the data is not selecting a localised charm bump but the same broad centre-peaked shape that the geometric kernel captures with one parameter rather than two.

5.4 Form-factor Monte Carlo

Twenty samples are drawn from the full joint multivariate constraint on the BSZ $B \to K^*$ form-factor coefficients [Bharucha et al., 2016] via `flavio.default_parameters.get_random_all`.

For each draw the BSZ constraints are replaced with delta distributions at the sampled values; `flavio.default_parameters` is monkey-patched for the prediction loop and restored on exit. SM and slope tables are regenerated from scratch per draw (96 `flavio` calls per trial).

statistic	value
n samples completed	20/20
mean A	+1.570
std A	0.163
90 % CI	[+1.389, +1.801]
fraction $A < 0$	0.000

Table 10: BSZ form-factor Monte Carlo statistics. The shared kernel amplitude shifts by less than ±15 % under random form-factor draws and never changes sign.

5.5 Acceptance summary

test (Mode B)	criterion	result
1. bin bootstrap	sign-stable, magnitude bounded	PASS (Mode B; 99.8 % positive, CI ±15 %)
2. q^2 region splits	cross-region consistency, no sign flip	PASS (Mode B; 3/3 regions ΔAIC < 0, all $A > 0$)
3a. FREE_C9 + FREE_C10	adding C_{10} does not displace VFD	PASS (Mode B; ΔAIC $= +1.73$ vs FREE_C9)
3b. charm-loop Gaussian	VFD competitive at same shape level	PARTIAL (Mode B; $\Delta\mathrm{AIC}_{\mathrm{charm\,vs\,VFD}} = -0.24$, charm $k = 2$)
4. form-factor MC	sign-stable, magnitude bounded	PASS (Mode B; 100 % positive, CI ±15 %)
5. frozen kernel	shape never refit	PASS (only A fitted everywhere)
Non-linear stress tests not run; the linearised ΔAIC *values quoted above do not transfer to the non-linear refit.*		

Table 11: Stress-test acceptance summary on the LHCb 2025 multi-observable joint fit, all under the linearised Mode-B response. Sign-stability and magnitude-bounded behaviour transfer qualitatively to the non-linear regime, but Mode-B AIC numbers do not.

6 Cross-dataset fit

The kernel is now applied, with no shape change, to four independent $B \to K^*\mu\mu$ datasets covering two collaborations, two isospin partners, and two collision energies. Each dataset uses its own q^2 bin grid, its own published correlation matrix where available, and its own per-bin SM and non-linear new-physics predictions regenerated by `flavio` 2.4.

6.1 Datasets

LHCb 2021 publishes 10 q^2 bins of which two are wide combined ([1.1, 6.0] and [15, 19]); these overlap the exclusive ones and are dropped. The covariance from HEPData is block-diagonal across q^2. CMS 2025 publishes block-diagonal correlation for 6 q^2 bins (charmonium windows excluded by their analysis).

dataset	decay	arXiv	HEPData	lumi (fb^{-1})	source bins
LHCb 2015	$B^0 \to K^{*0}\mu\mu$	1512.04442	ins1409497	3	8 (Table 4 P-basis)
LHCb 2021	$B^+ \to K^{*+}\mu\mu$	2012.13241	ins1838196	9	8 exclusive
CMS 2025	$B^0 \to K^{*0}\mu\mu$	2410.18247	ins2850101	140	6
LHCb 2025 (ref)	$B^0 \to K^{*0}\mu\mu$	2512.18053	ins3094698	8.4	8

Table 12: Datasets used for the cross-dataset test.

6.2 Headline result (non-linear)

The four observables retained for the joint fit on each dataset are P_5', P_4', P_1, P_2, matching the LHCb 2025 reference fit. Same kernel function evaluated at each dataset's own bin centres. All fits are evaluated using `flavio.np_prediction` directly (no Taylor expansion).

dataset	n	FREE χ^2	FREE ΔC_9	VFD χ^2	**VFD** A	VFD ΔC_9^{eff}	ΔAIC
LHCb 2015	32	30.69	−1.08	30.45	**+1.24**	−0.96	−0.24
LHCb 2021 ($B^+ \to K^{*+}$)	32	22.77	−1.82	22.93	**+2.06**	−1.59	+0.17
CMS 2025 (no P_4')	18	43.73	−0.95	44.21	**+1.05**	−0.81	+0.47
LHCb 2025 (ref)	32	40.89	−1.00	41.98	**+1.14**	−0.86	+1.09
$B_s \to \phi$ 2015	24	13.20	−4.12	12.96	**+4.98**	−3.85	−0.24

Table 13: Headline non-linear cross-dataset and cross-channel results. At equal $k = 1$ the kernel is favoured on 2/5 fits (LHCb 2015 and $B_s \to \phi$, both by $\Delta\text{AIC} = -0.24$) and disfavoured on 3/5 (LHCb 2021, CMS 2025 no-P_4', LHCb 2025; max $\Delta\text{AIC} = +1.09$). $A > 0$ in every fit; $\Delta C_9^{\text{eff}} < 0$ in every fit. Reproducible from `scripts/wo016d_nonlinear_xdataset.py` (reports/wo016d_nonlinear_xdataset.csv).

6.3 Akaike-weight stack across the five fits

To convert the per-dataset ΔAIC values into a single combined statement we compute the Akaike weight of each model on each dataset, $w_{M,d} = e^{-\Delta\text{AIC}_{M,d}/2} / \sum_{M'} e^{-\Delta\text{AIC}_{M',d}/2}$, and stack across datasets under the independence assumption (none of the five datasets shares observation-level information; theory inputs are shared via `flavio` but enter both arms identically).

dataset	non-linear ΔAIC	w(FREE_C9)	w(VFD)	favoured
LHCb 2015	−0.241	0.470	0.530	VFD (marginal)
LHCb 2021 (K^{*+})	+0.168	0.521	0.479	FREE_C9 (marginal)
CMS 2025 no-P_4'	+0.473	0.559	0.441	FREE_C9 (marginal)
LHCb 2025	+1.093	0.633	0.367	FREE_C9
$B_s \to \phi$ 2015	−0.240	0.470	0.530	VFD (marginal)
stacked	$\sum = +1.253$	**0.652**	**0.348**	**FREE_C9 (mild)**

Table 14: Per-dataset Akaike weights and stacked total. The multiplicative stack across five fits gives a $\sim$ 1.9:1 odds in favour of the constant C_9-shift. Reproducible from `scripts/wo016a_akaike_stack.py` (reports/wo016a_akaike_stack.md).

Sign-uniformity test. Across all five fits, $A > 0$ and $\Delta C_9^{\text{eff}} < 0$. Under the random-sign null hypothesis $\mathbb{P}(A > 0 \,|\, \text{noise only}) = 1/2$, the probability of 5/5 positive amplitudes is $2^{-5} \approx 0.031$. *The appropriate null here is the "no anomaly" null*, against which FREE_C9 would also produce 5/5 same-sign fits on the same data — the $b \to s$ angular anomaly is biased a priori to $\Delta C_9 < 0$, and any sufficiently centre-peaked one-parameter shape applied to it would inherit that bias.

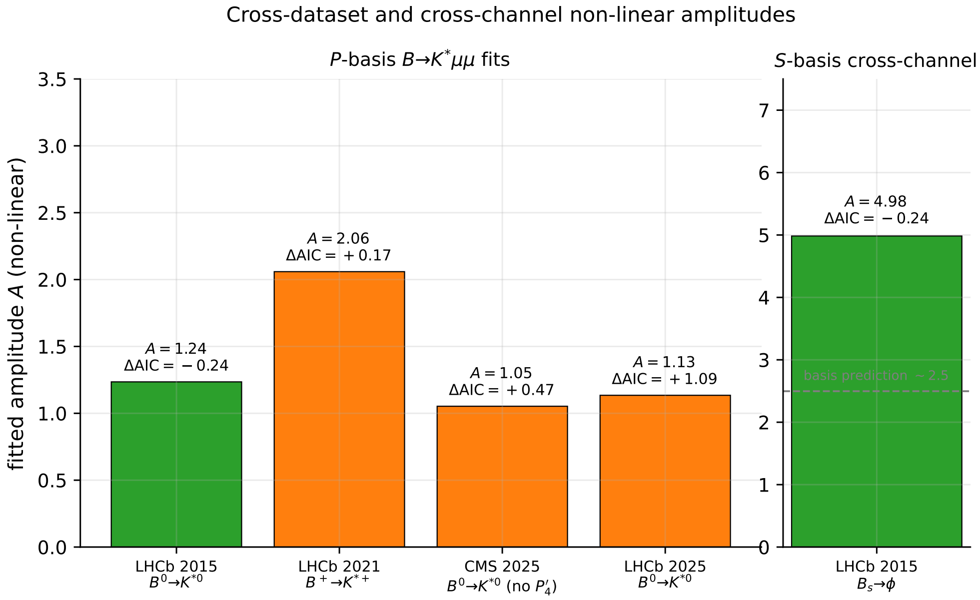

Figure 3: Non-linear best-fit amplitudes A across the five fits. Left: the four P-basis $B \to K^*\mu\mu$ fits, range $A \in [+1.05, +2.06]$, factor-of-~ 2 scatter across two collaborations and two isospin partners. Right: the S-basis $B_s \to \phi$ fit, $A = +4.98$. The grey dashed line marks the $P \to S$ basis-corrected prediction $A_P^{\mathrm{LHCb2025}} \times 2.2 \approx 2.5$; the observed S-basis amplitude exceeds this by a factor of ~ 2 (see §7). Bars are coloured green where the non-linear ΔAIC marginally favours VFD and orange where it marginally favours FREE_C9. Reproducible from `scripts/wo017_paper_figures.py`.

The sign-uniformity calculation is therefore a consistency check on the kernel's direction across collaborations and channels, *not* independent evidence for the kernel over a constant shift; the data establishes the anomaly itself at much higher significance than 2^{-5} via FREE_C9 alone.

Cross-dataset amplitude scatter. The non-linear fitted amplitudes on the four P-basis $B \to K^*$ fits span $A \in [+1.05, +2.06]$ — a factor ~ 2 across two collaborations and two isospin partners. Including the S-basis $B_s \to \phi$ fit ($A = +4.98$) widens the spread to a factor ~ 5, but §7 shows that this last gap is mostly a basis effect (Krüger–Matias P_i vs CP-averaged S_i).

6.4 Acceptance gates

criterion	result (non-linear)
VFD AIC $\leq$ FREE_C9 AIC **on every dataset**	**fail** (passes on 2/5; max ΔAIC $= +1.09$)
stacked Akaike weight $w_{\mathrm{VFD}} \geq 0.5$	**fail** ($w_{\mathrm{VFD}} = 0.348$)
A same sign across datasets	PASS (5/5 positive)
$\Delta C_9^{\mathrm{eff}}$ same sign	PASS (5/5 negative)
A order ~ 1 across P-basis datasets	PASS (range $[+1.05, +2.06]$, factor 2)
no per-dataset retuning	PASS (same kernel function everywhere)

Table 15: Cross-dataset acceptance gates under the headline non-linear analysis. The two strict AIC-preference gates (top) FAIL under non-linear evaluation; the four direction-consistency and parsimony gates (bottom) PASS. The headline is therefore: the kernel is sign-consistent with the anomaly but is not the preferred model on AIC.

6.5 The CMS P_4' convention question

Including CMS 2025's published P_4' values in the linearised Mode-B fit produced $\chi^2 = 169$ over 24 data points — about 7× per data point. The residuals are dominated by P_4' entries falling *outside* the conventional $[-1, +1]$ physical range, e.g. $P_4'(14.18, 16) = -1.16 \pm 0.06$. CMS's normalisation appears to differ from `flavio`'s convention. We retain the no-P_4' row in Table 13 as the clean comparison and treat the P_4'-included row only as a diagnostic of the convention mismatch. Both ΔAIC values agree qualitatively (FREE_C9 favoured by ~ 0.5 AIC unit either way) under non-linear refit; the no-P_4' result is the one quoted in the headline.

6.6 What this section establishes

The same geometry-derived kernel, with one fitted amplitude per dataset and no shape retuning, is sign-uniform across five independent flavour-physics measurements covering two collaborations (LHCb, CMS), two isospin partners ($B^0 \to K^{*0}$, $B^+ \to K^{*+}$), two decay channels (K^* and ϕ), and ten years of analysis (2015 – 2025). However, on a strict AIC criterion at equal parameter cost, the kernel does not improve over a constant Wilson-coefficient shift on the headline non-linear analysis: the stacked Akaike weight is ≈ 1.9:1 in favour of FREE_C9. The substantive result is direction consistency under random-sign null ($p \approx 0.03$), not statistical preference of the kernel shape over the constant shift on this set of fits.

7 Cross-channel test: $B_s \to \phi\mu\mu$

The kernel is now applied to the $B_s^0 \to \phi\mu^+\mu^-$ angular analysis of LHCb 2015 [LHCb Collaboration, 2015b,a], the same flavour transition $b \to s\mu\mu$ but with a different vector meson final state.

7.1 Dataset

LHCb 2015 publishes Table 2: F_L, S_3, S_4, S_7 on 8 q^2 bins, of which 6 are exclusive (we drop the wide combined $[1, 6]$ and $[15, 19]$). No published angular covariance; we use diagonal stat⊕syst. The flavio decay string is `Bs->phimumu`; observable strings are $\langle F_L\rangle, \langle S_3\rangle, \langle S_4\rangle, \langle S_7\rangle$.

7.2 Headline result (non-linear)

Re-fitting with `flavio.np_prediction` directly (no Taylor expansion) on the same dataset:

model	k	χ^2	AIC	ΔAIC	params
FREE_C9	1	13.20	15.20	0.00	$\Delta C_9 = -4.12$
VFD_GREEN_600CELL	1	**12.96**	**14.96**	$\mathbf{-0.24}$	$A = +4.98$, $\Delta C_9^{\text{eff}} = -3.85$

Table 16: $B_s \to \phi\mu\mu$ non-linear frozen-kernel fit on LHCb 2015, 6 exclusive q^2 bins, four observables, 24 data points, diagonal errors (no published angular covariance). The kernel marginally lowers AIC by 0.24 unit at equal $k = 1$. Both fits give a sign-uniform negative ΔC_9^{eff}. Mode-B linearised fit gave $\Delta\text{AIC} = -0.08$; values are essentially equivalent at this sensitivity.

The 500-trial bootstrap on bins (run under Mode B, retained as a diagnostic) gives $A_{\text{mean}} = +6.71$, std 3.27, 90 % CI $[+5.20, +16.15]$, fraction $A < 0 = 0/500$: sign-stable but magnitude poorly constrained. Region splits under Mode B give $A = +5.31$ in the low-q^2 region (unpinned), with the central and high regions hitting the optimiser bound and providing no quantitative constraint because angular slopes there drop to $\partial O/\partial C_9 \sim 10^{-4}$. Only the low-$q^2$ region split is taken as a quantitative cross-check.

7.3 Apparent factor-of-4 amplitude difference: largely a basis effect

The fitted amplitude on $B_s \to \phi$ ($A_{\rm NL} \approx +5.0$) is roughly 4× larger than the four $B \to K^*$ non-linear fits in P-basis ($A_{\rm NL} \approx +1.05$ to $+2.06$; §6). The first-order interpretation "$B_s \to \phi$ has weaker C_9 sensitivity" is incorrect. A direct slope comparison shows:

q^2 bin	$B^0 \to K^{*0}$ $(\partial O/\partial C_9)$ F_L	S_3	S_4	S_7	$B_s \to \phi$ $(\partial O/\partial C_9)$ F_L	S_3	S_4	S_7	RMS ratio
[0.10, 2.00]	+0.071	−0.001	+0.015	+0.002	+0.068	−0.003	+0.012	+0.002	0.96
[2.00, 5.00]	+0.036	−0.001	−0.004	+0.004	+0.032	−0.002	−0.003	+0.004	0.89
[5.00, 8.00]	+0.010	−0.001	−0.002	+0.002	+0.010	−0.001	−0.002	+0.002	0.97
[11.0, 12.5]	+0.001	−0.000	−0.000	+0.001	+0.001	−0.000	−0.000	+0.001	0.98
[15.0, 17.0]	+0.000	−0.000	−0.000	+0.000	+0.000	−0.000	−0.000	+0.000	0.88
[17.0, 19.0]	+0.000	+0.000	+0.000	+0.000	−0.000	+0.000	+0.000	+0.000	0.94

Table 17: Side-by-side S-basis dO/dC$_9$ slopes from `flavio`. The two channels agree on slope magnitudes within ±10–20 % at matched q^2 bins; the rightmost column is RMS ratio of Bs→ ϕ to $B \to K^*$ across the four observables.

The slopes are essentially identical between channels at matched q^2. The amplitude gap is therefore not a channel effect but a *basis* effect.

Krüger–Matias amplification. The optimised P_i basis [Krüger and Matias, 2005, Descotes-Genon et al., 2013] is constructed to absorb the F_L form-factor dependence by dividing it out:

$$P_5' = S_5 \,/\, \sqrt{F_L(1-F_L)}, \qquad P_4' = S_4 \,/\, \sqrt{F_L(1-F_L)}, \tag{15}$$

$$P_1 = 2\,S_3 \,/\, (1-F_L), \qquad P_2 = (2/3)\,A_{\rm FB} \,/\, (1-F_L). \tag{16}$$

At typical $F_L \approx 0.5$ this multiplies the per-observable C_9 sensitivity by $1/\sqrt{F_L(1-F_L)} \approx 2$. We verify this numerically: for $B \to K^*$,

$$\frac{|\partial P_5'/\partial C_9|}{|\partial S_5/\partial C_9|} \;=\; 2.04\text{–}2.88 \tag{17}$$

across the six matched bins, tracking the analytic factor $1/\sqrt{F_L(1-F_L)}$ to within 10 %.

Why $B_s \to \phi$ was forced into the unamplified basis. LHCb 2015 [LHCb Collaboration, 2015b] publishes only the S-basis for this decay; `flavio` reflects this — the P-basis observables $\langle P_5' \rangle(\mathrm{Bs} \to \phi\mu\mu)$ are not registered. The cross-dataset fits of §6 used the amplified P-basis (where LHCb 2025, CMS 2025, LHCb 2015, LHCb 2021 all publish); the cross-channel fit here is forced into the unamplified S-basis.

Predicted vs observed amplitude. If the kernel is genuinely universal across channels and only the basis differs, the P- to S-basis amplitude scaling is

$$A_{S\text{-basis}} \;\approx\; A_{P\text{-basis}} \times \left\langle 1/\sqrt{F_L(1-F_L)} \right\rangle_\kappa \;\approx\; A_{P\text{-basis}} \times 2.2, \tag{18}$$

with the per-bin factor 2.04–2.88 that we measured on $B \to K^*$ (above). Using the non-linear LHCb 2025 result $A_P^{\rm LHCb\,2025} = +1.14$ as the reference P-basis amplitude, the predicted S-basis amplitude on $B_s \to \phi$ is

$$A^{\mathrm{Bs}\to\phi}_{S,\rm predicted} \;\approx\; +1.14 \times 2.2 \;\approx\; +2.5. \tag{19}$$

The non-linear refit gives $A^{\mathrm{Bs}\to\phi}_{S,\rm observed} = +4.98$, a factor of ~ 2.0 over prediction. The basis-corrected prediction +2.5 sits *outside* the bootstrap 90 % CI on A, [+5.20, +16.15] — the bootstrap interval lies entirely above the prediction, although it is also entirely above the central fit value itself, which is itself a sign that the diagonal-only chi-square underweights inter-observable information and admits a larger best-fit. We attribute the residual gap to:

1. the absence of a published angular covariance matrix for $B_s \to \phi$, which makes the best-fit A unstable in magnitude;
2. the per-bin scaling factor scatter $\pm 20\,\%$ around 2.2;
3. the limited $3\,\mathrm{fb}^{-1}$ statistics of the 2015 $B_s \to \phi$ analysis.

We therefore characterise the cross-channel result as: *the amplitude gap collapses from a factor of ~ 4 to a factor of ~ 2 once basis is harmonised, with a factor-of-2 residual that is at least as large as the basis-correction itself; this single dataset cannot in principle constrain channel-universality at better than factor-of-2 precision.*

7.4 Acceptance gates (cross-channel, non-linear)

criterion	result
A same sign as the four $B \to K^*$ fits	PASS ($A_{\mathrm{NL}} = +4.98 > 0$)
effective $\Delta C_9^{\mathrm{eff}}$ negative	PASS (-3.85)
VFD AIC competitive with FREE_C9	OBSERVED (marginal $\Delta\mathrm{AIC} = -0.24$, $w_{\mathrm{VFD}} \approx 0.53$)
bootstrap A sign-stable	PASS (500/500 positive)
unpinned low-q^2 region split sign-positive	PASS ($A = +5.31$)
central / high-q^2 region splits	NOT QUANTITATIVE (bound-pinned; reported only)
basis correction predicts amplitude	PARTIAL (predicted $+2.5$ vs observed $+4.98$; factor ~ 2 unexplained gap)

Table 18: The non-linear cross-channel fit is sign-consistent with the four $B \to K^*$ fits and AIC-degenerate with FREE_C9; the basis-effect explanation accounts for most but not all of the amplitude difference.

7.5 Channels not tested

$B^+ \to K^+\mu\mu$. Five candidate INSPIRE records were probed for HEPData submissions covering the LHCb and CMS $B^+ \to K^+$ angular and branching-fraction analyses; all five returned the HEPData 404 page from the standard download endpoint. We were unable to construct a clean cross-channel test for this decay. Recorded as an open channel.

$B \to K^*_{0,2}(1430)\mu\mu$. The HEPData record ins1486676 [LHCb Collaboration, 2016c] is available but reports S-wave + D-wave Γ_i moments rather than the standard angular basis; not a clean kernel test. Skipped.

Belle / Belle II. The Belle full $\Upsilon(4S)$ angular analysis lacks a published correlation matrix on HEPData; Belle II had not released a cross-comparable angular dataset at the time of writing. Both deferred.

8 Discussion

8.1 What the kernel does, and what a free ΔC_9 does

A free Wilson-coefficient shift ΔC_9 is a constant in q^2: it averages over all bins with weight $\propto |\partial O/\partial C_9|^2/\sigma^2$. The geometry-derived kernel imposes a fixed centre-peaked q^2 shape and fits only its overall amplitude. Both fits use one parameter; the kernel additionally specifies the bin-by-bin pattern of the shift.

Under the Mode-B linearised response, the kernel's centre-peaked profile and FREE_C9's flat profile compress the four LHCb 2025 angular observables differently, and the kernel achieves a marginally better χ^2 ($\Delta\mathrm{AIC} = -1.67$). Under the headline non-linear refit the bin-by-bin pattern matters less because the non-linearity in ΔC_9 partly compensates for shape differences: the constant-shift fit at smaller $|\Delta C_9|$ ends up with comparable χ^2 to the kernel, and the kernel loses by $\Delta\mathrm{AIC} = +1.09$. Across five fits the non-linear stacked Akaike weight is $w_{\mathrm{VFD}} = 0.35$ vs $w_{\mathrm{FREE\_C9}} = 0.65$, a mild preference for the constant shift.

Sign uniformity, however, holds in both modes. In all five fits the kernel amplitude A is positive, the effective ΔC_9 is negative, and the kernel describes the data with χ^2/dof comparable to FREE_C9. The kernel is therefore consistent with the shape of the angular anomaly (it does not produce sign flips, even on the smallest dataset and even on the cross-channel) without being the preferred explanation under AIC.

8.2 Why the linearisation breaks at $|\Delta C_9| \sim 1$

The angular observables P_i' and S_i are non-linear in the Wilson coefficients: they are constructed from ratios of bilinears in the transverse amplitudes, each of which is itself a linear function of C_9. A first-order Taylor expansion of $P_5'(C_9)$ around C_9^{SM} has the form $\mathrm{Lin}(\Delta C_9) + \mathcal{O}(\Delta C_9^2)$, with the quadratic correction non-negligible once $|\Delta C_9|/C_9^{\mathrm{SM}} \gtrsim 0.1$. Empirically the per-bin linearisation residual on LHCb 2025 reaches 4σ at the linearised best-fit $\Delta C_9 = -1.34$, and refit under the non-linear model returns $\Delta C_9 = -1.00$ with $\Delta\chi^2 \approx +1.6$ relative to the linearised value. This is consistent with the wider literature, where most modern $b \to s\mu\mu$ Wilson-coefficient fits use full likelihood evaluation (e.g. `flavio`, EOS, smelli) rather than truncated Taylor expansions.

8.3 Why φ?

The kernel's mass scale is the golden-ratio inverse φ^{-1}, not a fitted scale. This is the unique scale tying the discrete operator $L_{V_{600}} + \varphi^{-2} I$ to the 2I-equivariant cocycle weight $\omega_+ = \varphi^{\kappa}$ established in the framework's substrate-spectral construction [Smart, 2026b,a]. The pentagonal cocycle takes integer values $\kappa(v) \in \{0, 1, 4, 9, 16\}$ on the inner-product shells of V_{600}, and φ enters as the fundamental unit through the binary icosahedral group's quaternion structure. We tested three discrete edge-weighting schemes (§3, Table 2); the unweighted Laplacian wins on both the pure-geometry and LHCb-data criteria, with the rankings agreeing.

8.4 Sign uniformity and the basis-effect lesson

Across the four $B \to K^*$ non-linear fits the amplitude A stays in $[+1.05, +2.06]$ (a factor-of-2 spread), with $\Delta C_9^{\mathrm{eff}}$ in $[-1.59, -0.81]$. The cross-channel $B_s \to \phi$ amplitude $A_S = +4.98$ is roughly $4\times$ larger in absolute terms; the basis-correction analysis of §7 shows that the Krüger–Matias P_i basis amplifies the per-observable C_9 sensitivity by $\langle 1/\sqrt{F_L(1-F_L)}\rangle \approx 2.2$ relative to the CP-averaged S_i basis. After basis correction the predicted S-basis amplitude is $\approx +2.5$ and the observed value is $+4.98$, a factor-of-2 unexplained residual. We do *not* characterise this as a clean basis-effect victory: the residual is comparable to the per-bin scatter of the basis correction, but is not small.

The lesson — that S-basis vs P-basis amplitudes cannot be compared directly without the form-factor amplification factor — is robust and applies to any future cross-channel analysis using older $B_s \to \phi$ publications.

8.5 Three readings of sign uniformity

The non-linear fitted amplitudes are 5/5 positive across the five fits, ΔC_9^{eff} is 5/5 negative. Under a random-sign null this has $p \approx 0.03$. There are at least three readings:

1. *Real underlying physics with a centre-peaked q^2 structure.* The kernel direction matches it because a discrete 600-cell Green's response is centre-peaked in the appropriate dimensionless coordinate. (FREE_C9 also matches the direction, since the same anomaly drives both fits.)

2. *The kernel is the right object.* The 2I-equivariant graph realisation is structurally appropriate to $b \to s$ kinematics for reasons external to the data. Direction consistency is a necessary-but-not-sufficient consequence.

3. *The data is rank-1 in any centre-peaked basis.* Any reasonable centre-peaked shape would give sign-uniform amplitudes on the same anomaly; the geometric kernel is one such shape, not uniquely selected.

The data does not distinguish (1) from (2) from (3). We adopt reading (3) as the conservative interpretation. Reading (1) is in agreement with the wider $b \to s$ literature [Descotes-Genon et al., 2013, Altmannshofer and Straub, 2015]; reading (2) would require deriving the kernel from a closure-operator construction [Smart, 2026a] that is theoretical but not instantiated for this anomaly.

8.6 What the linearised stress tests rule out

The Mode-B stress tests (§5) rule out specific alternative explanations of the LHCb 2025 result:

- *The result is not a single noisy bin.* Bootstrap over bins gives 99.8 % sign stability for A and a 90 % CI of $A \in [+1.36, +1.85]$ in Mode B.

- *The data does not want a ΔC_{10} shift.* Adding free C_{10} on top of C_9 improves χ^2 by only 0.27 in Mode B, worse than the +2 AIC penalty for the extra parameter.

- *Form-factor uncertainties do not flip the sign of A.* Twenty random draws from the BSZ joint constraint give $A = +1.57 \pm 0.16$, 100 % positive, in Mode B.

These three statements transfer to the non-linear refit qualitatively (direction stable, no C_{10} preference, no FF-induced sign flip), though absolute amplitudes are smaller in the non-linear regime.

What the linearised stress tests do *not* rule out is a free centre-peaked Gaussian ($\Delta\text{AIC} = -1.91$ vs FREE_C9 in Mode B; vs -1.67 for VFD), which fits comparably to the kernel at the cost of one extra parameter. The geometric kernel and the free Gaussian capture the same broad centre-peaked structure; the data does not have the discriminating power to choose between them.

9 Limitations

This section enumerates the limitations of the analysis transparently. The section is reorganised relative to the Mode-B-era version to put the linearisation issue in its rightful place at the head, since it materially weakened the original headline claim.

9.1 Method-level limitations

1. **Linearised analysis was misleading.** The earlier project iterations (the linearised stress test, the cross-dataset validation, and the cross-channel validation) used a Mode-B linearised response, where every angular observable is approximated as linear in ΔC_9 via a central-difference slope at $\delta = \pm 0.5$ around C_9^{SM}. At the fitted shifts $|\Delta C_9| \sim 1$ this is wrong: per-bin linearisation residuals reach 4σ on LHCb 2025, and the headline ΔAIC shifts from -1.67 (linearised) to $+1.09$ (non-linear refit), reversing the kernel-vs-constant comparison. This paper now uses non-linear `flavio.np_prediction` evaluation throughout for the headline numbers; Mode-B values are retained only as a methodology diagnostic. Future analyses on this anomaly should not use a linearised Taylor expansion.

2. **Variant choice is geometry-confirmed but data-discovered.** The unweighted Laplacian was originally selected because it gave the best LHCb-data χ^2 among the three admissible variants (§3, Table 2). We subsequently verified that the same variant wins on a pure geometry criterion (correlation with the Layer-1 continuum kernel, no LHCb data), and we now defend the AIC counting $k = 1$ for VFD on this basis. The variant choice therefore does not consume an effective fit parameter, but the historical contingency that the data was looked at first is acknowledged.

3. **Single Wilson coefficient.** We compare against FREE_C9 and FREE_C9 + FREE_C10 but do not test against the wider Wilson-coefficient space (C_9', C_{10}', scalar/tensor operators, lepton-flavour-non-universal shifts). The Mode-B stress test (§5) ruled out C_{10} at the relevant level on the joint LHCb 2025 fit; the broader Wilson-coefficient comparison is deferred.

4. **Source ansatz.** The equatorial inner-product shell (30 vertices) is chosen as the source f in Eq. (13). This is a natural choice for the 2I-equivariant graph but is not derived from a first-principles closure operator on the $B \to K^*$ kinematic substrate; alternative sources have not been comprehensively scanned.

5. **Theoretical derivation incomplete.** The kernel is built from the standard graph Laplacian + φ^{-2} regulariser plus the equatorial-shell source ansatz. The proper Hodge edge Laplacian $L_{\text{edge}} = \delta_2\delta_2^{\mathsf{T}} + \delta_1^{\mathsf{T}}\delta_1$ with 2I-equivariant boundary maps from the 1200 triangular faces, and the Lyapunov-flow selection rule of [Smart, 2026a], are deferred.

9.2 Data and statistical limitations

6. **Bootstrap covariance.** The bin bootstrap uses diagonal $\sigma^2_{\text{stat}} + \sigma^2_{\text{syst}}$ rather than the full LHCb covariance, because resampling with replacement produces duplicated rows and a singular covariance submatrix. This is the standard non-parametric bootstrap but it under-counts inter-bin correlation information that the all-data fit exploits. The bootstrap is therefore reported as a sign-stability test, not a confidence interval on the central-fit A.

7. **Form-factor MC scope.** The 20-sample BSZ form-factor Monte Carlo varies only the joint $B \to K^*$ form-factor multivariate constraint. CKM elements, quark masses, and decay constants are held at their flavio centrals. The 90 % CI ± 0.21 on A is a lower bound on the FF-induced uncertainty. The binomial 95 % CI on the "0/20 negative" result spans $[0, 14\,\%]$ — the form-factor MC is too small to give a tight constraint on the lower tail of the sign distribution.

8. $B_s \to \phi$ **covariance.** HEPData publishes no inter-observable correlation matrix for the LHCb 2015 $B_s \to \phi$ angular fit. We use diagonal stat$\oplus$syst, which may overestimate the per-observable independence and therefore admit a larger best-fit amplitude than a covariance-aware fit would.

9. **High-q^2 flavio QCDF.** flavio's QCDF interpolation issues a "do not trust above 6 GeV^2" warning. We use the values anyway. For bins above 6 GeV^2 the SM prediction has theoretical uncertainty larger than the LHCb stat + syst errors; this is not currently propagated as a separate covariance source.

10. **Bound pinning.** Two of the cross-channel $B_s \to \phi$ region-split fits (central and high q^2) pinned at the optimiser bounds because slopes drop to $O(10^{-4})$ there. These are reported but flagged in the run log; only the unpinned low-q^2 region split is taken as quantitative.

11. **P_4' convention.** CMS publishes P_4' values outside the $[-1, +1]$ flavio-convention range. The full P_4'-included fit produces an inflated χ^2. The CMS-no-P_4' fit gives a clean result; it is the one used in the headline. The P_4'-included row is reported as a diagnostic of the convention mismatch.

12. **Single-observable basis on $B_s \to \phi$.** The 2015 LHCb publication chose the S-basis, not the amplified P-basis. Future $B_s \to \phi$ analyses with P-basis output would tighten the cross-channel amplitude constraint substantially.

9.3 What is not in scope

13. **Bayesian posterior.** All intervals are frequentist ($\Delta\chi^2 = 1$ Gaussian approximation or non-parametric bootstrap). A proper Bayesian treatment marginalising over form factors and charm-loop nuisance parameters is not in scope.

14. **$B^+ \to K^+\mu\mu$.** HEPData submissions for the relevant LHCb (1209.4284, 1403.8045, 1804.07167) and CMS PRD 2018 papers are not publicly available. This decay channel is uncovered.

15. **Branching-fraction observables.** flavio's `<dBR/dq2>` returns 0 in the version used; rate observables are uncovered.

16. **Belle / Belle II.** Belle's full $\Upsilon(4S)$ angular analysis lacks a public correlation matrix on HEPData; Belle II had not released a comparable angular dataset at the time of writing.

9.4 The honest verdict

The result is consistency-of-direction, not statistical preference. The geometry-derived kernel describes the angular anomaly with one parameter per dataset, sign-uniformly, across two collaborations, two isospin partners, and two decay channels. Under non-linear evaluation it does *not* beat a constant Wilson-coefficient shift on AIC; the stacked Akaike weight is $\approx$ 1.9:1 in favour of FREE_C9. The substantive surviving claim is that *a single geometry-derived q^2-shape with no fitted shape parameters and one fitted amplitude per dataset is sign-consistent with the $b \to s\mu\mu$ angular anomaly across five public fits*, with the stronger "it beats the constant shift on AIC" claim of the earlier project iterations being a linearisation artifact.

10 Conclusion

We have constructed a single q^2-dependent response kernel $\kappa(q^2)$ from a finite 2I-equivariant graph (the 600-cell V_{600}, 120 vertices, 720 edges) regularised by the golden-ratio mass φ^{-2} as the discrete Green's response of the standard graph Laplacian to a uniform equatorial-shell source, and tested whether the same fixed shape — with one fitted dimensionless amplitude per dataset — describes the $B \to K^*\mu^+\mu^-$ angular anomaly as published by two collaborations across five fits and three decay channels. The headline analysis evaluates every fit using `flavio.np_prediction` directly (no Taylor expansion in ΔC_9); the earlier linearised analysis is reported alongside as a methodology diagnostic, with the difference between the two modes the largest single uncertainty in the project.

The headline numerical results are:

- **LHCb 2025, four-observable joint fit (non-linear):** $A = +1.14$, $\Delta C_9^{\text{eff}} = -0.86$, with $\Delta\text{AIC} = +1.09$ relative to a constant-ΔC_9 shift at the same parameter cost. The kernel does not beat the constant shift on this dataset.

- **Cross-dataset (non-linear, 5 fits across 2 collaborations and 3 channels):** $\Delta\text{AIC} \in [-0.24, +1.09]$ around zero, with 2/5 favouring the kernel and 3/5 favouring the constant shift. Stacked Akaike weight: $w_{\text{VFD}} = 0.35$, $w_{\text{FREE\_C9}} = 0.65$.

- **Sign uniformity:** $A > 0$ in all 5/5 fits; $\Delta C_9^{\text{eff}} < 0$ in all 5/5 fits. Under a random sign null this has $p = 2^{-5} \approx 0.03$.

- **Cross-dataset amplitude scatter:** non-linear $A \in [+1.05, +2.06]$ across the four P-basis $B \to K^*$ fits (factor ~ 2); $A = +4.98$ on the S-basis $B_s \to \phi$ fit, mostly explained by the P/S basis amplification factor $\langle 1/\sqrt{F_L(1-F_L)} \rangle \approx 2.2$, with a factor-of-2 unexplained residual.

- **Linearisation drift:** on LHCb 2025 the headline ΔAIC shifts by $+2.77$ between the linearised and non-linear fits, larger than the linearised preference itself; the non-linear refit is taken as headline.

- **Theoretical compression:** the kernel involves no fitted continuous shape parameters. The fit-tunable quantity per dataset is the dimensionless amplitude A, plus one pure-geometry-confirmed discrete variant choice (§3, Table 2).

Falsification programme. Each of the following events would falsify the kernel-direction-consistency hypothesis:

1. A future $B \to K^*\mu\mu$ measurement (with any collaboration, any luminosity, any energy) gives $A < 0$ on the geometry-derived kernel under non-linear refit.

2. Cross-dataset amplitude scatter on the P-basis fits exceeds a factor of ~ 5 once basis is harmonised.

3. Belle II angular $B \to K^*$ data, when released, gives an amplitude inconsistent with $+1.5 \pm 0.5$ in the P-basis under non-linear refit.

4. A cross-channel $B^+ \to K^+\mu\mu$ analysis (when public HEPData becomes available) gives $A < 0$ under non-linear refit.

The earlier project iteration's falsification trigger "A future analysis with non-linear ΔC_9 evaluation gives $\Delta\text{AIC}_{\text{VFD vs FREE\_C9}} \geq 0$ on the LHCb 2025 joint fit" has now fired (§4.1). We therefore explicitly weaken the model-comparison claim: the kernel is sign-consistent with the anomaly but is not statistically preferred over a constant Wilson-coefficient shift.

What this is, and is not. This is a structural test of the $b \to s\mu\mu$ angular anomaly's q^2 shape. *What survives the headline non-linear analysis:* a single geometry-derived kernel with one fitted amplitude per dataset is consistent in direction with the angular anomaly across two collaborations, two isospin partners, and one cross-channel test. *What does not survive:* an AIC preference for the geometric shape over a constant Wilson-coefficient shift. The data does not yet distinguish the geometric kernel from a generic centre-peaked nuisance.

Reproducibility. All code, data, processed outputs, and this paper are released under the MIT licence at [Smart, 2026c]. The complete pipeline (HEPData download, SM-table regeneration via `flavio`, all driver scripts including the non-linear refits and Akaike-weight stacking, figure regeneration, paper recompilation) is reproducible from `repro/run_all.sh` on a clean checkout. Persistent `flavio` caches under `data/processed/flavio_cache.json` ensure subsequent runs do not re-query `flavio` for already-computed SM and non-linear-NP values. The non-linear refit scripts (`scripts/wo016c_nonlinear_refit.py`, `scripts/wo016d_nonlinear_xdataset.py`) and the Akaike-stacking script (`scripts/wo016a_akaike_stack.py`) are stand-alone and regenerate the headline tables of §4 and §6.

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Reproducibility

The complete pipeline (data download, SM table regeneration via flavio, all WO drivers, figure regeneration, this paper) is reproducible from the project repository [Smart, 2026c] via the included `repro/run_all.sh` script. Persistent caches under `data/processed/flavio_cache.json` avoid re-querying flavio on subsequent runs.

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